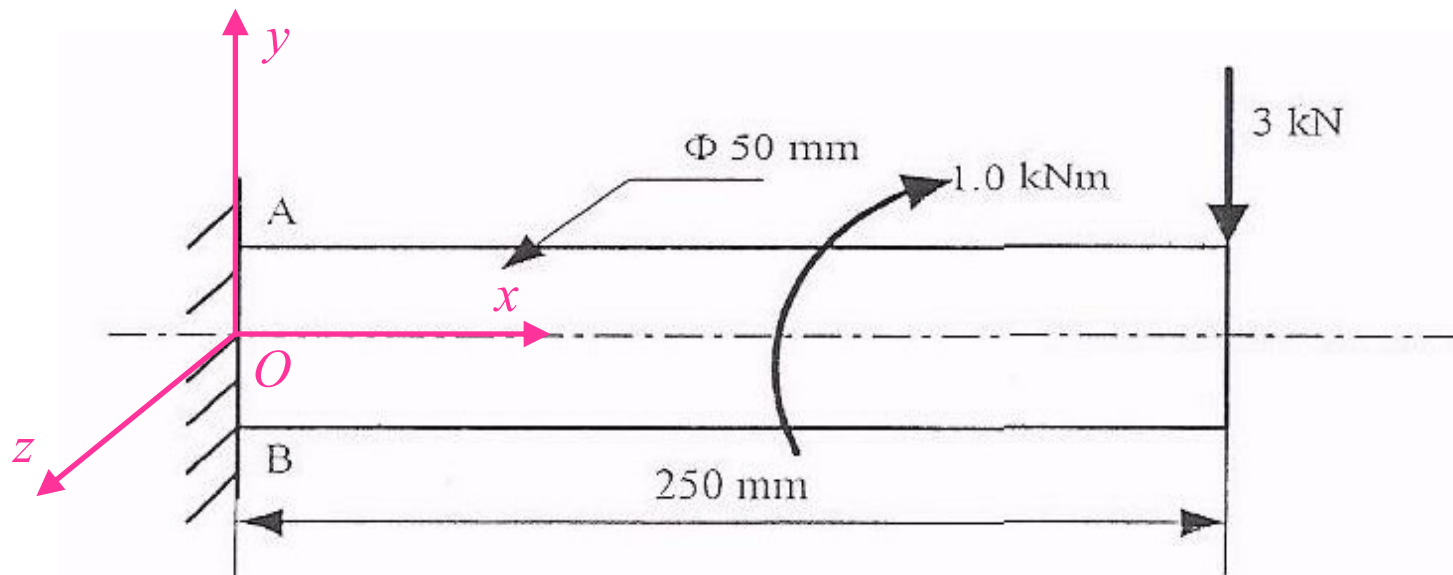


1. A machine member (50 mm diameter by 250 mm long) is supported at one end as a cantilever. Determine the principal stresses and the absolute maximum shear stress at point *A* and *B*. (30' in total)



Solutions to Extra Exercise #1:

At point A:

Due to the concentrated force 3 kN, the resultant moment at the section AB is:
 $M = 750 \text{ N} \cdot \text{m}$.

This moment will induce **tensile** normal stress (along x -direction):

$$\sigma_x = + \frac{M \cdot y}{I} = \frac{750 \times 0.025}{\frac{1}{4} \pi \cdot 0.025^4} = 61.1 \text{ MPa}$$

The torque will induce shear stress (along z -direction):

$$\tau_{xz} = \frac{Tc}{J} = \frac{1000 \times 0.025}{\frac{1}{2} \pi \cdot 0.025^4} = 40.7 \text{ MPa}$$

Thus, this is a plane stress problem for xz -plane.

Solutions to Extra Exercise #1, cont'd:

Given $\sigma_x = 61.1$ MPa, $\sigma_z = 0$, $\tau_{xz} = 40.7$ MPa, the center of the Mohr's circle C is located at the point $C(30.55, 0)$ and the reference point $A(61.1, 40.7)$ are labeled. Then, the radius of the circle is $R = \sqrt{(61.1 - 30.55)^2 + 40.7^2} = 50.89$, and the circle is constructed.

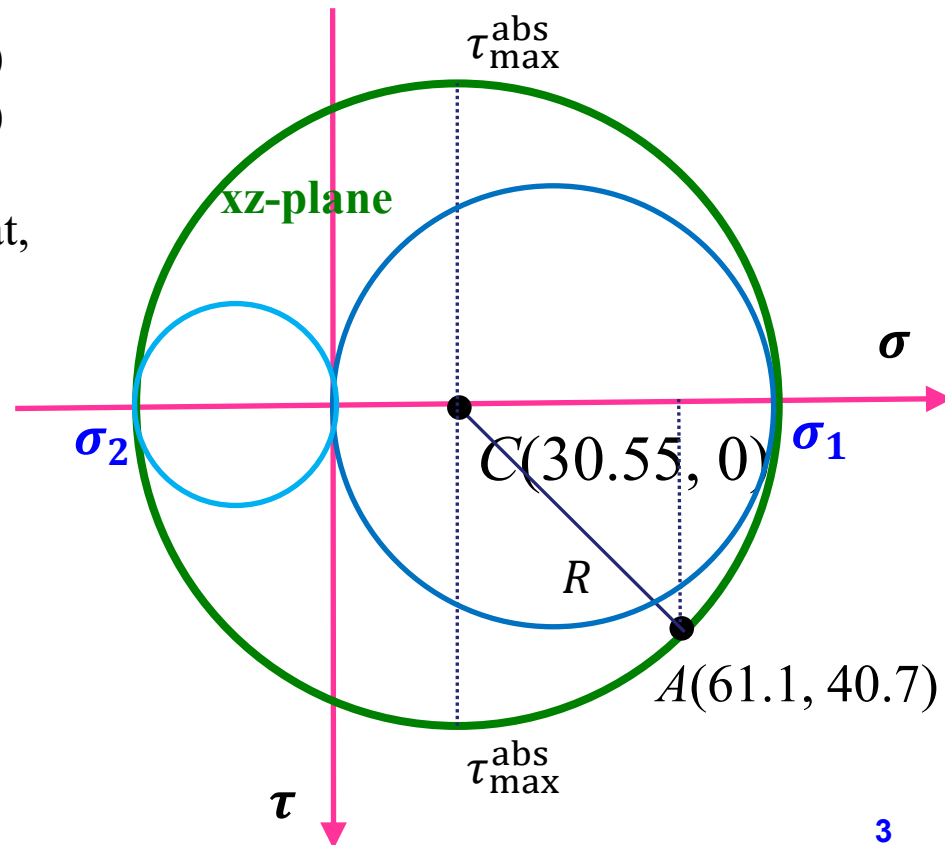
The principal stresses are:

$$\sigma_1 = 30.55 + 50.89 = \mathbf{81.44 \text{ MPa}} \quad (\text{Ans.})$$

$$\sigma_2 = 30.55 - 50.89 = \mathbf{-20.34 \text{ MPa}} \quad (\text{Ans.})$$

By drawing three Mohr's circles, we know that, the absolute maximum shear stress occurs in the xz -plane:

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1 - \sigma_2}{2} = \mathbf{50.89 \text{ MPa}} \quad (\text{Ans.})$$



Solutions to Extra Exercise #1, cont'd:

At point B:

Due to the concentrated force 3 kN, the resultant moment at the section AB is:
 $M = 750 \text{ N} \cdot \text{m}$.

This moment will induce **compressive** normal stress:

$$\sigma_x = -\frac{M \cdot y}{I} = -\frac{750 \times 0.025}{\frac{1}{4}\pi \cdot 0.025^4} = -61.1 \text{ MPa}$$

The torque will induce shear stress:

$$\tau_{xz} = \frac{Tc}{J} = \frac{1000 \times 0.025}{\frac{1}{2}\pi \cdot 0.025^4} = 40.7 \text{ MPa}$$

Again, this is a plane stress problem for xz -plane.

Solutions to Extra Exercise #1, cont'd:

Given $\sigma_x = -61.1$ MPa, $\sigma_z = 0$, $\tau_{xz} = 40.7$ MPa, the center of the Mohr's circle C is located at the point $C(-30.55, 0)$ and the reference point $A(-61.1, 40.7)$ are labeled. Then, the radius of the circle is $R = \sqrt{(-61.1 + 30.55)^2 + 40.7^2} = 50.89$, and the circle is constructed.

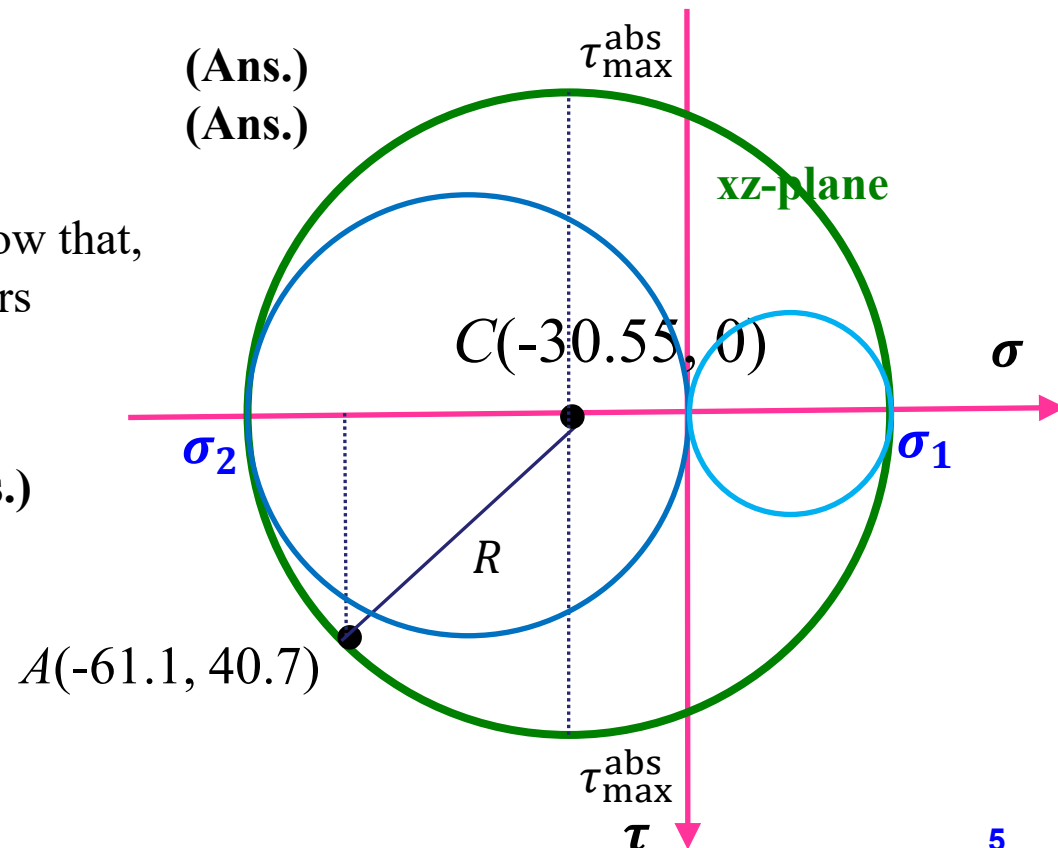
The principal stresses are:

$$\sigma_1 = -30.55 + 50.89 = \mathbf{20.34 \text{ MPa}}$$

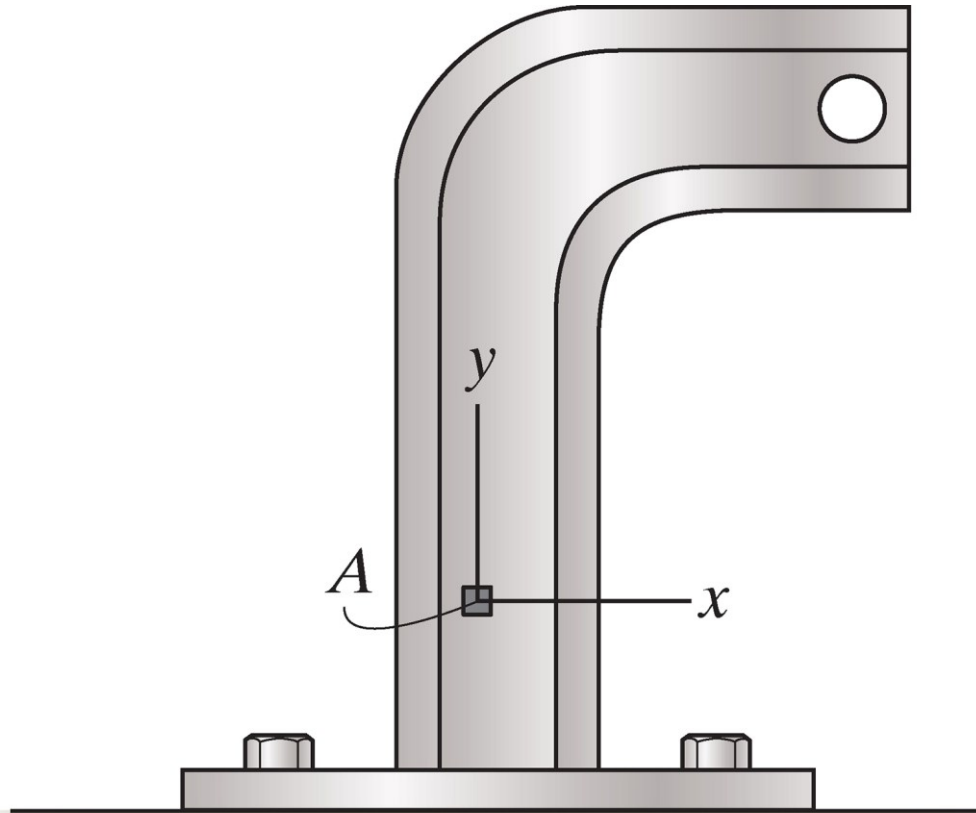
$$\sigma_2 = -30.55 - 50.89 = \mathbf{-81.44 \text{ MPa}}$$

By drawing three Mohr's circles, we know that, the absolute maximum shear stress occurs in the xz -plane:

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1 - \sigma_2}{2} = \mathbf{50.89 \text{ MPa}} \quad (\text{Ans.})$$



2. The strain at point A on the bracket has components $\varepsilon_x = 300 \times 10^{-6}$, $\varepsilon_y = 550 \times 10^{-6}$, $\gamma_{xy} = -650 \times 10^{-6}$, $\varepsilon_z = 0$. Determine (a) the principal strains at point A in the x - y plane, (b) the maximum shear strain in the x - y plane, and (c) the absolute maximum shear strain. (**20'** in total)



Solutions to Extra Exercise #2:

This is a typical plane strain problem. Given $\varepsilon_x = 300 \times 10^{-6}$, $\varepsilon_y = 550 \times 10^{-6}$, $\gamma_{xy}/2 = -325 \times 10^{-6}$, the Mohr's circle for the plane strain can be constructed.

$$\text{Center } C\left(\frac{\varepsilon_x + \varepsilon_y}{2}, 0\right) \rightarrow C(425 \times 10^{-6}, 0)$$

$$\text{Reference point } A(300 \times 10^{-6}, -325 \times 10^{-6})$$

$$\text{Radius: } R = \sqrt{(425 - 300)^2 + 325^2} \times 10^{-6} = 348.2 \times 10^{-6}$$

(a) Principal strains:

$$\varepsilon_1 = (425 + 348.2) \times 10^{-6} = \mathbf{7.732 \times 10^{-4}} \quad (\text{Ans.})$$

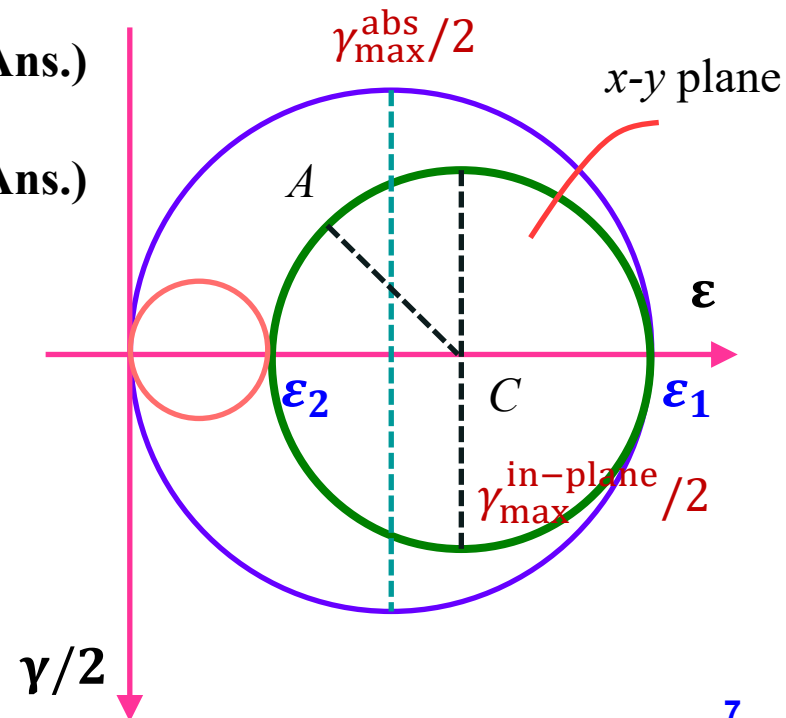
$$\varepsilon_2 = (425 - 348.2) \times 10^{-6} = \mathbf{7.68 \times 10^{-5}} \quad (\text{Ans.})$$

(b) Maximum shear strain in the x-y plane:

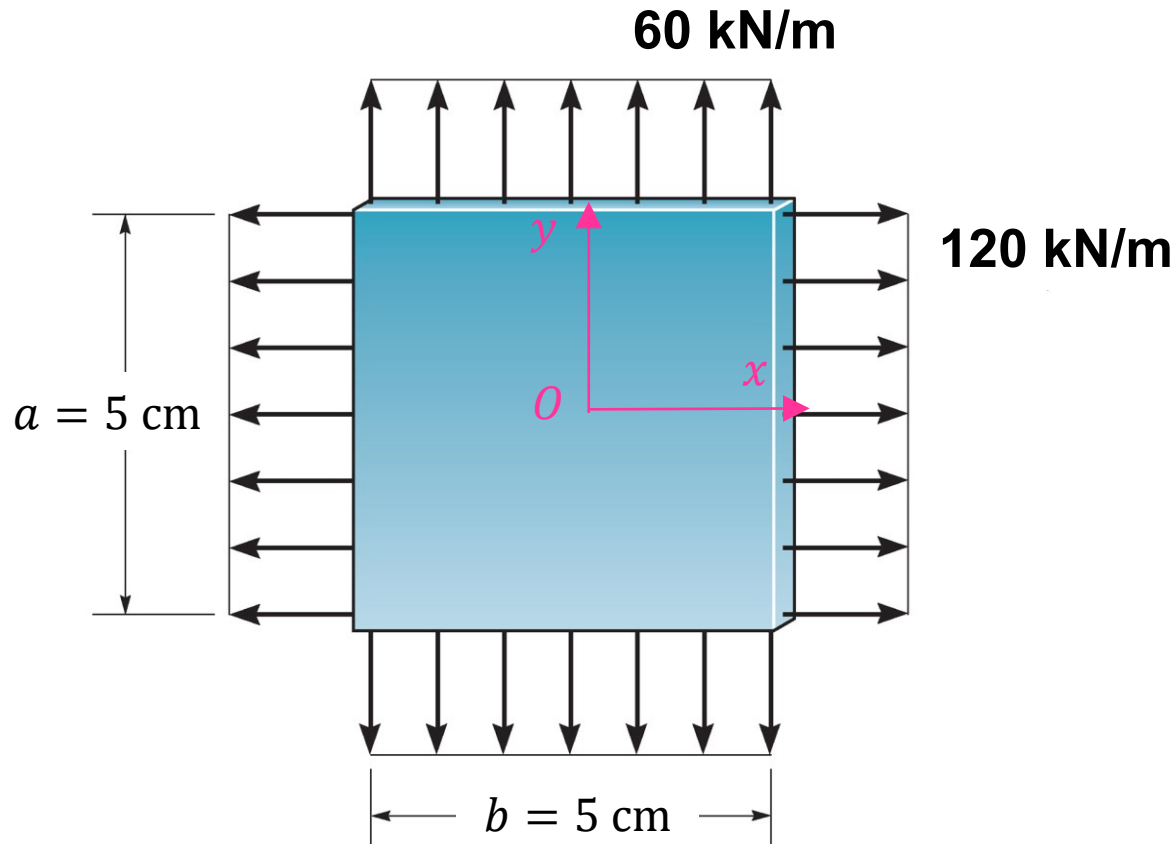
$$\gamma_{\max}^{\text{in-plane}} = 2R = \mathbf{6.964 \times 10^{-4}} \quad (\text{Ans.})$$

(c) The absolute maximum shear strain:

$$\gamma_{\max}^{\text{abs}} = \varepsilon_1 = \mathbf{7.732 \times 10^{-4}} \quad (\text{Ans.})$$



3. A uniform plate is subjected to the edge loads. If the plate is originally square and has dimensions of $a = b = 5$ cm, and a thickness of $t = 0.5$ cm, determine its new dimensions a' , b' , and t' after the loads are applied. Take the Young's modulus $E = 10$ GPa and the Poisson's ratio $\nu = 0.20$. (20' in total)



Solutions to Extra Exercise #3:

The two normal stresses in the xy-plane of the plate are:

$$\sigma_x = \frac{120 \times 10^3 \times 0.05}{0.05 \times 0.5 \times 10^{-2}} = 24 \text{ MPa}$$

$$\sigma_y = \frac{60 \times 10^3 \times 0.05}{0.05 \times 0.5 \times 10^{-2}} = 12 \text{ MPa}$$

According to the generalized Hooke's law, the strain in three directions are:

$$\varepsilon_x = \frac{1}{E} [\sigma_x - \nu \cdot (\sigma_y + \sigma_z)] = \frac{24 \times 10^6 - 0.2 \times (12 \times 10^6)}{10 \times 10^9} = 2.16 \times 10^{-3}$$

$$\varepsilon_y = \frac{1}{E} [\sigma_y - \nu \cdot (\sigma_x + \sigma_z)] = \frac{12 \times 10^6 - 0.2 \times 24 \times 10^6}{10 \times 10^9} = 7.2 \times 10^{-4}$$

$$\varepsilon_z = \frac{1}{E} [\sigma_z - \nu \cdot (\sigma_x + \sigma_y)] = \frac{-0.2 \times (24 + 12) \times 10^6}{10 \times 10^9} = -7.2 \times 10^{-4}$$

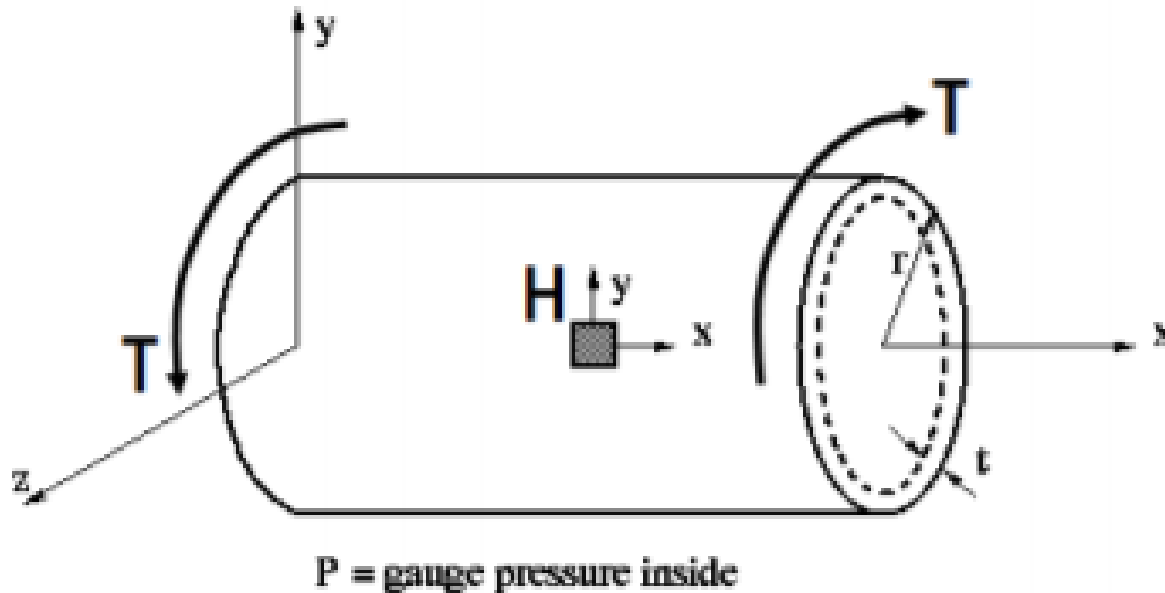
Then, the new dimension of the plate is:

$$a' = a \cdot (1 + \varepsilon_y) = \mathbf{5.0036 \text{ cm}} \quad (\text{Ans.})$$

$$b' = b \cdot (1 + \varepsilon_x) = \mathbf{5.011 \text{ cm}} \quad (\text{Ans.})$$

$$t' = t \cdot (1 + \varepsilon_z) = \mathbf{0.4996 \text{ cm}} \quad (\text{Ans.})$$

4. A cylindrical pressure vessel has an outer diameter of 100 cm and wall thickness of 0.32 cm. The vessel has an internal gas pressure of 0.7 MPa and an end-torque of $T = 700 \text{ N} \cdot \text{m}$. Sketch the state of stress for the point H on the surface of the vessel and determine the principal stresses and the **absolute maximum shear strain** at this point. Take the Young's modulus $E=160 \text{ GPa}$ and the Poisson's ratio $\nu=0.25$. (30' in total)



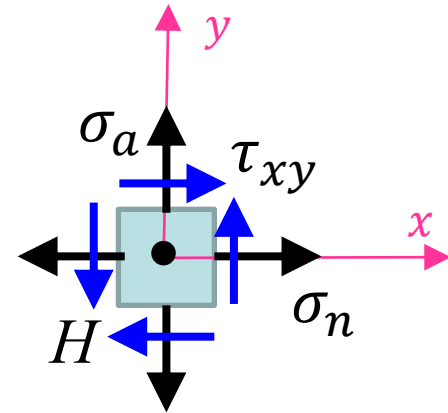
Solutions to Extra Exercise #4:

The stress state of point H is sketched in the figure, which can be obtained by adding additional shear stress (induced by the torque) to the original state induced by the internal pressure.

$$\sigma_x = \sigma_n = \frac{pr}{2t} = \frac{0.7 \times 10^6 \times 0.5}{2 \times 0.0032} = 54.7 \text{ MPa}$$

$$\sigma_y = \sigma_a = \frac{pr}{t} = \frac{0.7 \times 10^6 \times 0.5}{0.0032} = 109.4 \text{ Mpa}$$

$$\tau = \frac{Tc}{J} = \frac{700 \times 0.5}{\frac{\pi}{2} \times (0.5^4 - 0.4968^4)} = 0.14 \text{ MPa}$$



Solutions to Extra Exercise #4, cont'd:

All stresses occur in xy -plane. Thus, this is a plane-stress problem for xy -plane.

Given $\sigma_x = 54.7$ MPa, $\sigma_y = 109.4$ MPa, $\tau_{xy} = 0.14$ MPa, the center of the Mohr's circle C is located at the point $C(82.05, 0)$ and the reference point $A(54.7, 0.14)$ are labeled.

Then, the radius of the circle is $R = \sqrt{(82.05 - 54.7)^2 + 0.14^2} = 27.35$, and the circle is constructed.

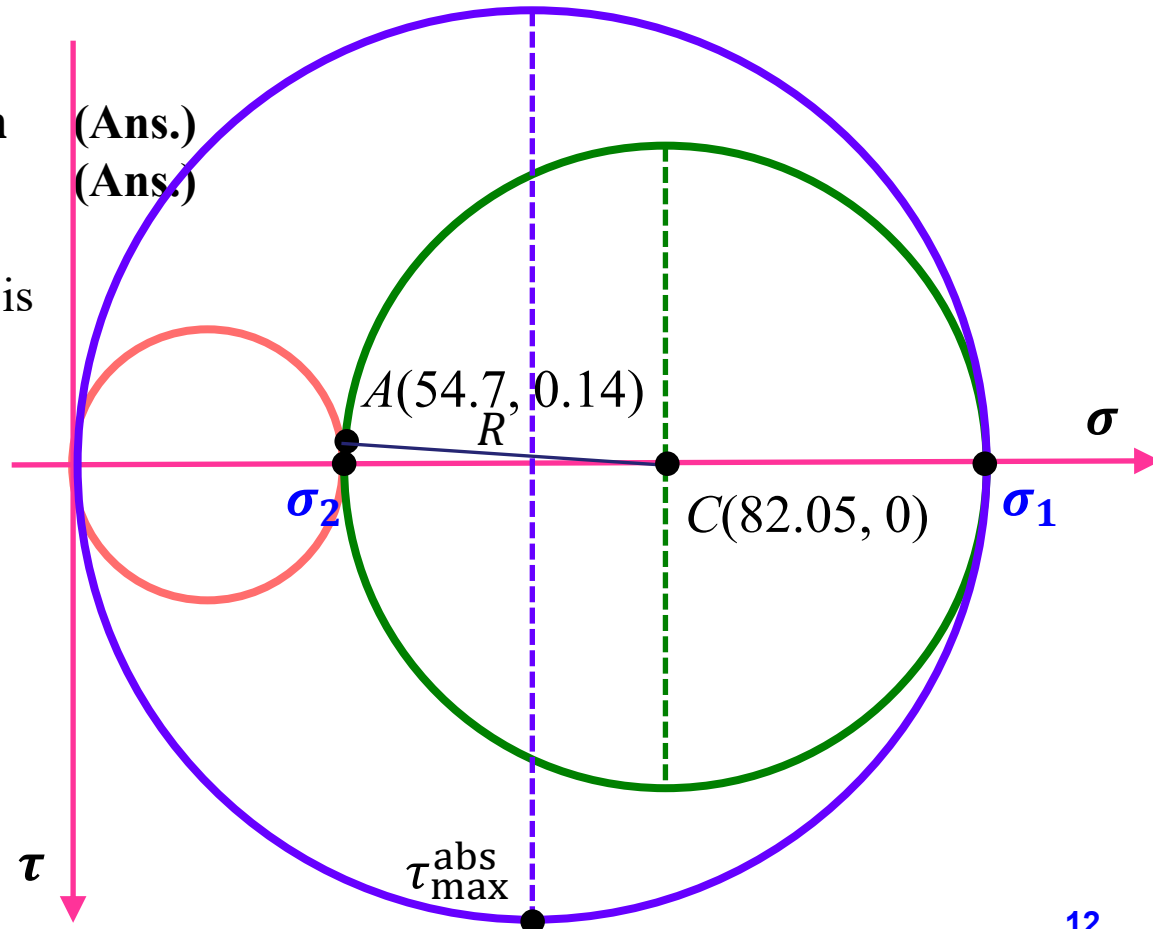
The principal stresses are:

$$\sigma_1 = 82.05 + 27.35 = \mathbf{109.4 \text{ MPa}} \quad (\text{Ans.})$$

$$\sigma_2 = 82.05 - 27.35 = \mathbf{54.7 \text{ MPa}} \quad (\text{Ans.})$$

The absolute maximum shear stress is

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1}{2} = 54.7 \text{ MPa}$$



Solutions to Extra Exercise #4, cont'd:

The absolute maximum shear strain can be determined using the Hooke's law for shear.

First, we calculate the shear modulus by the material property relationship

$$G = \frac{E}{2(1 + \nu)} = \frac{160}{2 \times (1 + 0.25)} = 64 \text{ Gpa}$$

Finally, the absolute maximum shear strain is

$$\gamma_{\max}^{\text{abs}} = \frac{\tau_{\max}^{\text{abs}}}{G} = \frac{54.7 \times 10^6}{64 \times 10^9} = \mathbf{8.5469 \times 10^{-4} \text{ rad}} \quad \mathbf{(Ans.)}$$